POG_DENSE: Expanding the Candidate Score

From $\theta(c \mid S)$ to training bundles and item descriptions

Take a candidate item c and a nonempty partially filled set S, with $c \notin S$. Let $\mathcal{I}$ be the catalogue, $n = |\mathcal{I}|$, and $B \in \{0,1\}^{m \times n}$ the training table: row e is a past bundle, and $B_{ei} = 1$ if it contains item i. The catalogue size is supplied in `count.json`; these training rows are the memberships supplied in `bi_train.txt`. Define the observed frequency $f_i = \sum_e B_{ei}$. Our task is to compute $\theta(c \mid S)$ and rank candidates by this score. This note follows the POG_DENSE formulation with full item features. We use the default uncapped known set S. The item categories and title, image, and other description vectors are additional observed inputs; they cannot be recovered from B alone. Their construction is not assumed to follow from bundle membership. The supplied clothing label of item i is denoted by $\ell(i)$. A title vector, an image vector, and a description vector each contain numerical coordinates supplied with the dataset; they are not counts computed from B. Their upstream encoders are not specified by the scoring procedure, so no particular encoder is assumed in the equations below. We specify exactly how the supplied vectors are transformed after they are available. Define $[x]_+ = \max(x, 0)$, $\varepsilon_8 = 10^{-8}$, $\varepsilon_9 = 10^{-9}$, and $\varepsilon_{12} = 10^{-12}$. $\mathbf{1}[A]$ is one when statement A is true and zero otherwise; $\|v\|_2$ is vector length, and $v \cdot w$ is the sum of products of corresponding entries. The numerical constants below specify this version of the method. Their roles can be explained, but their exact values are design choices rather than mathematical consequences.

1 Start with the score: two explicitly weighted estimates

Before giving the score, define $Y(c \mid S)$ as the combined evidence, $P(c \mid S)$ as an estimate drawn mainly from categories and item descriptions, $s_0 \geq 0$ as its scale, and $\kappa \geq 0$ as the strength of the fallback. These are fixed after selection on held-out examples. The final score is

$$\theta(c \mid S) = \begin{cases} Y(c \mid S), & \kappa = 0, \\ \dfrac{f_c}{f_c + \kappa} Y(c \mid S) + \dfrac{\kappa}{f_c + \kappa} s_0 P(c \mid S), & \kappa > 0. \end{cases} \tag{1.1}$$

Why this term? The assumption behind this mixture is that more appearances make history-based evidence more dependable. Frequency alone does not prove that assumption for every item. A candidate with no appearances receives $s_0 P$ when the fallback is enabled. For example, if $f_c = \kappa$, both estimates receive weight $1/2$. The result is a ranking score. The next question is what goes into the two estimates Y and P.

1.1 Expand the combined evidence Y

Let $p(c \mid S)$ denote a weighted sum of training-bundle matches. Let $\delta(c \mid S)$ denote a second sum that includes only training bundles containing at least two members of S. Let L be a fixed item-to-item matrix built from B. Section 3 gives their complete definitions. Here the important distinction is the exact pair of raw calculations used for each source:

Source h	Raw score r_h^0	Raw score r_h
bundle	p_c	$p_c + \delta_c$
propagated	$(Lp)_c$	$(Lp)_c + (L\delta)_c$
Each remaining source	r_h	r_h

Here p, δ are column vectors, with $p_c = p(c \mid S)$ and $\delta_c = \delta(c \mid S)$. Let $\mathcal{H}$ be the following ordered list:

$$(\text{cat}, \text{soft}, \text{text}, \text{image}, \text{bridge}, \text{bundle}, \text{propagated}, \text{pop}, \text{cooc}, \text{pmi}, \text{svd}).$$

These are labels for the formulas expanded below, not additional assumptions. Let $w_h \geq 0$ be the selected weight of source h. Apply the same standardization rule $\mathcal{Z}_S$ separately to each raw column, as defined in Section 2: $z_h^0 = \mathcal{Z}_S[r_h^0]$ and $z_h = \mathcal{Z}_S[r_h]$. Let $\alpha \in [0, 1]$ be a mixing coefficient, $\lambda_R \geq 0$ an agreement coefficient, and z_R the standardized pair-of-known-items score defined in Section 7. Then

$$Y_0(c \mid S) = \sum_{h \in \mathcal{H}} w_h z_h^0(c \mid S), \tag{1.2}$$

$$Y_1(c \mid S) = \sum_{h \in \mathcal{H}} w_h z_h(c \mid S), \tag{1.3}$$

$$Y(c \mid S) = (1 - \alpha) Y_0(c \mid S) + \alpha Y_1(c \mid S) + \lambda_R z_R(c \mid S). \tag{1.4}$$

Where each source comes from. Here a source means one explicit raw scoring rule. The following table identifies its inputs and the intermediate object used to compare a candidate with S; the subsequent sections define every intermediate object.

Source	Input observations	Construction used for scoring
bundle	Training bundles B	Weighted overlap p, optionally plus joint-overlap correction δ
propagated	B and bundle scores	Item connection matrix L applied to p or $p+\delta$
pop	Training appearances	Logarithm of frequency f_c
cooc	Training bundles B	Item-pair occurrence counts C
pmi	Pair counts C	Positive log-ratio table M relative to pair-endpoint marginals
svd	Table M	Similarity of projected association profiles U
text, image	Supplied vectors and clothing labels	Similarity after subtracting category means
bridge	All supplied vector modalities, clothing labels, and B	Residual descriptors combined with descriptors of connected items
cat	Clothing labels and B	Category-pair counts and category repetition rates
soft	Supplied vectors and B	MiniBatch K-means clusters, soft cluster memberships, and within-bundle cluster-pair counts

The separately weighted score r_R combines candidate-to-seed connections from B with residual title-vector similarities and seed category labels. Neither its name nor any source label denotes an additional observed dataset.

Why this term? The only difference between Y_0 and Y_1 is the explicit δ contribution in the two rows of the table. This is a definition of this model: descriptions, category scores and the other counts are held fixed during this comparison. It is not a claim that only these two sources can describe relationships among several items. For instance, r_R also uses pairs of known items, but it is a separate term with its own coefficient λ_R. Restricting α to $[0, 1]$ makes $1 - \alpha$ and α nonnegative weights adding to one; values outside this interval would extrapolate rather than average. Requiring $\lambda_R \geq 0$ keeps a positive relative agreement score from being penalized through a negative coefficient. These are modeling restrictions. At $\alpha = 0$, the two rows use their left-hand formulas; at $\alpha = 1$, they use their right-hand formulas. Intermediate α mixes the separately standardized scores. It does not replace p by $p + \alpha\delta$ before standardization.

1.2 Expand the fallback P

Define $\mathcal{Q} = \{\text{cat}, \text{soft}, \text{text}, \text{image}, \text{bridge}\}$ restricted to available sources, and let $\mathcal{A} = \{h \in \mathcal{H} : w_h > 0\}$ after removal of rejected weights. The prior-source set is defined precisely by

$$\mathcal{P} = \begin{cases} \mathcal{Q} \cap \mathcal{A}, & \mathcal{Q} \cap \mathcal{A} \neq \varnothing, \\ \mathcal{Q}, & \mathcal{Q} \cap \mathcal{A} = \varnothing. \end{cases}$$

This is a prescribed choice of prior sources, not a set forced by a theorem. For $h \in \mathcal{P}$, let $\widetilde{w}_h = w_h$ when its positive weight was retained, and $\widetilde{w}_h = 0.5$ otherwise. An omitted source has $w_h = 0$ in Y; the value 0.5 in P is a separate default rule. Let z_{pop} denote the standardized popularity score. Then

$$\boxed{P(c \mid S) = \sum_{h \in \mathcal{P}} \widetilde{w}_h z_h(c \mid S) + \lambda_R z_R(c \mid S) + w_{\text{pop}} z_{\text{pop}}(c \mid S).} \tag{1.5}$$

Why this term? For an item with $f_c = 0$, its row in $B^\top B$ and in the associated training-only item matrices is zero. Such rows cannot provide item-specific positive shared-history support. Categories and supplied feature vectors can still distinguish that item, so $\mathcal{Q}$ selects those kinds of sources. This is the motivation for this restriction, not a proof that this particular restriction is best. Bridge and category tables still use training history; the fallback is not independent of B. Selected popularity and agreement also enter this estimate. The scale s_0 makes its overall standard deviation comparable to that of Y. We now expand the standardized scores z_h used in both equations.

2 Expand z_h: put each source on a common scale

Let $r_h(i \mid S)$ be the raw score of source h for catalogue item i. Define $\varepsilon_9 = 10^{-9}$, the catalogue mean μ_h, and sample standard deviation σ_h by

$$\mu_h(S) = \frac{1}{n} \sum_{i \in \mathcal{I}} r_h(i \mid S), \tag{2.1}$$

$$\sigma_h(S) = \sqrt{\frac{1}{n-1} \sum_{i \in \mathcal{I}} (r_h(i \mid S) - \mu_h(S))^2}, \tag{2.2}$$

$$\boxed{z_h(c \mid S)} \boxed{= \frac{r_h(c \mid S) - \mu_h(S)}{\sigma_h(S) + \varepsilon_9}.} \tag{2.3}$$

Denote this column operation by $\mathcal{Z}_S[r_h]$. Apply it separately to r_h^0 and to the agreement score r_R. The sums include the known items; they are excluded only when producing suggestions. For a fixed request these statistics depend on S and the already constructed score sources, across all catalogue items. They do not use which candidates are hidden correct completions, or the scores of other test requests.

Why this term? A shared-history count and an image similarity have different units. This conversion expresses each as above or below its own catalogue average. A constant raw score becomes zero and carries no distinction between candidates. In particular, normalizing $p + \delta$ is different from separately normalizing p and δ and adding them. The inclusion of known items and the divisor $n - 1$ are fixed conventions of this formulation; neither is required by the definition of a ranking task.

The remaining task is to define each raw source in the ordered list $\mathcal{H}$ already given. We expand the bundle and propagated scores first, since they also define the connections used by the other parts of the score.

3 Expand the two training-bundle formulas

3.1 Direct bundle matches: $p(c \mid S)$ and the extra overlap $\delta(c \mid S)$

Define bundle size $d_e = \sum_i B_{ei}$, known-item weight $u_s = (f_s + 5)^{-1/2}$, and bundle weight $v_e = (d_e + 1)^{-1/2}$. Let o_e be the weighted overlap with the known items, and k_e their unweighted overlap count:

$$o_e(S) = v_e \sum_{s \in S} B_{es} u_s, \qquad k_e(S) = \sum_{s \in S} B_{es}, \tag{3.1}$$

$$p(c \mid S) = \sum_e B_{ec} o_e(S), \qquad \delta(c \mid S) = \sum_e B_{ec} o_e(S)\, 1.5[k_e(S) - 1]_+. \tag{3.2}$$

Here $[x]_+ = \max(x, 0)$. The two raw versions entering Y_0 and Y_1 are

$$\boxed{r^0_{\text{bundle}}(c \mid S) = p(c \mid S), \qquad r_{\text{bundle}}(c \mid S) = p(c \mid S) + \delta(c \mid S).} \tag{3.3}$$

Why this term? p adds evidence from past bundles containing the candidate and known items. For example, the factor $(f_s + 5)^{-1/2}$ decreases when a known item's count increases, and $(d_e + 1)^{-1/2}$ decreases as a bundle becomes larger. The offsets 5 and 1 prevent singular weights and soften the adjustment. The exponent $1/2$ and these offsets are fixed choices; many alternative decreasing functions would also be mathematically valid. More explicitly, a bundle with $k_e = 0$ or 1 contributes zero to δ; $k_e = 2$ contributes $1.5 B_{ec} o_e$; $k_e = 3$ contributes $3 B_{ec} o_e$. This condition concerns how many members of S occur in a particular training row, as specified by $k_e(S)$. The candidate must also occur in that row, through B_{ec}. Thus $\delta = 0$ for $|S| = 1$, and also whenever no training row contains two known items. The factor 1.5 and the linear increase in $k_e - 1$ specify a chosen strength and shape; they are not derived optimal values.

3.2 Spread history: apply the table L to the bundle scores

Let $D = \text{diag}((f_i + 5)^{-1/2})$ and $E = \text{diag}((d_e + 1)^{-1})$ be diagonal tables, and let $\text{off}(A)$ mean A with its diagonal replaced by zero. Define

$$L = \text{off}(DB^\top EBD). \tag{3.4}$$

For a vector q with one entry per item, $(Lq)_c = \sum_i L_{ci} q_i$. Write p and δ for the vectors of the scores just defined. Then

$$\boxed{r^0_{\text{propagated}}(c \mid S) = (Lp)_c, \qquad r_{\text{propagated}}(c \mid S) = (Lp)_c + (L\delta)_c.} \tag{3.5}$$

Why this term? A candidate can benefit from its connection to another item that already has strong support. L passes evidence through shared training bundles while excluding direct self-transfer. Entry by entry, for $c \neq i$,

$$L_{ci} = \frac{1}{\sqrt{f_c + 5}\sqrt{f_i + 5}} \sum_e \frac{B_{ec} B_{ei}}{d_e + 1}, \qquad L_{cc} = 0.$$

Thus $(Lp)_c$ adds p_i from each other item i, weighted by an explicitly defined shared-bundle connection. This is what "propagated" means here. A previously unseen item has a zero row in L and receives no such transfer.

4 Expand the other history sources

4.1 Popularity, shared counts, and unexpectedly frequent pairs

Define $C = \mathrm{off}(B^\top B)$, $g_i = \sum_j C_{ij} + \varepsilon_9$, and $Z_C = \sum_{i,j} C_{ij}$. Introduce $\varepsilon_{12} = 10^{-12}$ and define

$$M_{ij} = \begin{cases} [\log(C_{ij} Z_C / (g_i g_j) + \varepsilon_{12})]_+, & C_{ij} > 0, \\ 0, & C_{ij} = 0. \end{cases} \tag{4.1}$$

The raw scores are

$$\boxed{r_{\mathrm{pop}}(c \mid S) = \log(1 + f_c), \quad r_{\mathrm{cooc}}(c \mid S) = \sum_{s \in S} C_{cs}, \quad r_{\mathrm{pmi}}(c \mid S) = \sum_{s \in S} M_{cs}.} \tag{4.2}$$

Why this term? Popularity gives a general frequency preference. Shared counts measure direct pair evidence. M instead asks whether the pair is strong relative to the overall shared counts of its two items. Taking a logarithm compresses large ratios; taking the positive part discards below-expected associations from this source rather than assigning them a negative raw penalty. These are distinct views of the same training table. Ignoring the numerical stabilizers, sample an ordered pair of distinct items from all ordered within-bundle pairs. Its empirical probability is C_{ij}/Z_C, and the probability that its first endpoint is i is g_i/Z_C. The ratio inside the logarithm compares the observed pair probability with the product of the endpoint probabilities. This logarithm is pointwise mutual information (PMI); clipping negative values gives positive PMI (PPMI). The label pmi in $\mathcal{H}$ refers to this clipped table. In particular, $g_i - \varepsilon_9 = \sum_e B_{ei}(d_e - 1)$ counts pair endpoints, whereas f_i counts bundles containing i. These are different marginals when bundle sizes vary.

4.2 A short summary of the pair table

This source uses singular value decomposition (SVD) of the PPMI table M; it does not decompose B, the raw pair-count table C, or item descriptions. An exact SVD writes $M = A\Sigma V^\top$, where columns of A and V are orthonormal and the diagonal entries of Σ are nonnegative singular values ordered from largest to smallest. The right singular vectors are the columns of V; each has one coordinate per item and specifies a direction in the space of association profiles M_i. These directions are estimated numerically with a randomized truncated SVD and pseudo-random seed zero.

Symmetry and eigenvectors. The count table C is symmetric, and the denominator $g_i g_j$ is unchanged by swapping i, j. Hence $M = M^\top$. Symmetry permits an eigendecomposition

$$M = Q\Lambda Q^\top, \qquad Q^\top Q = I, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n). \tag{4.3}$$

Here Q contains eigenvectors and λ_j are signed eigenvalues. Even though every entry of M is nonnegative, its eigenvalues can be negative. An SVD can be chosen with right vectors $V = Q$, singular values $\sigma_j = |\lambda_j|$, and left vectors equal to $\mathrm{sign}(\lambda_j)Q_j$ for nonzero eigenvalues, after ordering columns by decreasing $|\lambda_j|$. Thus the retained directions can be understood as eigenvectors selected by eigenvalue magnitude, rather than by the largest signed eigenvalues.

More precisely, any exact right singular vector v satisfies $M^2 v = \sigma^2 v$. If opposite eigenvalues $+\lambda, -\lambda$ share the same magnitude, a numerical SVD may return mixtures within that singular subspace; an individual returned vector then need not be an eigenvector of M itself. The eigenbasis description is a valid choice, whereas the projected-score definition below also covers such mixtures and numerical approximations.

Set $r = \min(128, n-2)$. The limit 128 is a chosen summary width, not an intrinsic rank of the data. Let $V_r \in \mathbb{R}^{n\times r}$ contain the leading r right singular vectors of M: orthogonal directions that, for an exact decomposition, give its best rank-r approximation in squared error. A numerical approximation to these directions is sufficient to define the fitted feature table. Introduce $\varepsilon_8 = 10^{-8}$ and the row-normalization rule $\mathcal{N}(F)_i = F_i/(\|F_i\|_2+\varepsilon_8)$. Define the projected item table $Z = MV_r$ and the normalized item table $U = \mathcal{N}(Z)$; if M is zero, use $U = 0$. The score is

$$\boxed{r_{\mathrm{svd}}(c \mid S) = U_c \cdot \left(\frac{1}{|S|}\sum_{s\in S} U_s\right).} \tag{4.4}$$

Why normalize after an orthonormal decomposition? The columns of V_r are directions in $\mathbb{R}^n$ and satisfy $V_r^\top V_r = I_r$. The scored vectors are instead the *rows* Z_i, each in $\mathbb{R}^r$. In an exact decomposition $Z = A_r\Sigma_r$, so

$$Z_{ij} = \sigma_j(A_r)_{ij}, \qquad \|Z_i\|_2^2 = \sum_{j=1}^{r} \sigma_j^2(A_r)_{ij}^2. \tag{4.5}$$

Nothing in column orthonormality makes this row length equal to one or equal across items. Define $b_i = \|Z_i\|_2$. Then an unnormalized comparison has the form $Z_c \cdot Z_s = b_c b_s \cos\phi_{cs}$ for nonzero rows, where ϕ_{cs} is their angle. Row normalization changes it to

$$U_c \cdot U_s = \frac{Z_c \cdot Z_s}{(b_c+\varepsilon_8)(b_s+\varepsilon_8)}. \tag{4.6}$$

This compares the direction of the two projected association profiles, removing their row magnitudes apart from the stabilizer. It is a chosen cosine-based scoring rule, not an additional orthogonalization. It does not set all singular values to one: their relative weighting within each row is retained. The later catalogue standardization is different again; it rescales an entire score column and cannot in general remove these item-specific row lengths. In this notation A_r denotes the left singular vectors; U denotes the normalized item table used for scoring.

Why this term? This summarizes many shared-history relationships in a shorter vector per item. The dot product compares the candidate's summary with the mean summary of the known items. It supplements the full pair table rather than replacing all history scores with a short vector. More explicitly, before row normalization its item-comparison matrix is $MV_rV_r^\top M^\top$, which compares projected rows of M. At full rank this is $MM^\top$, with entries $\sum_j M_{cj}M_{sj}$: the candidate and seed receive support for associating with the same other items. Thus the source can connect items with similar association partners even when their own direct pair count is zero. It is different from looking up a rank-r reconstruction of the direct association M_{cs}.

5 Expand the text, image, and bridged-description sources

Let R^t, R^i, R^d be the supplied title, image and description vector tables, respectively. They are read from `title_emb.npy`, `content_feature.pt` and `description_feature.pt`. For each available modality ν, define its normalized table by

$$F_i^\nu = \frac{R_i^\nu}{\|R_i^\nu\|_2 + \varepsilon_8}. \tag{5.1}$$

A zero input row stays zero. These supplied arrays are the starting observations; this scoring procedure does not generate their upstream embeddings. The superscripts t, i, d name modalities; the subscript in F_i^ν identifies the catalogue item. Each modality may have a different number of coordinates, but all rows within that modality use the same coordinate system. Let $\ell(i)$ be item

i's clothing category. For modality $\nu \in \{t, i, d\}$, define the category average $\bar{F}^{\nu}_{a}$ and the remaining feature D^{ν}:

$$\bar{F}^{\nu}_{a} = \frac{\sum_{i:\ell(i)=a} F^{\nu}_{i}}{|\{i : \ell(i) = a\}|}, \qquad D^{\nu}_{i} = \frac{F^{\nu}_{i} - \bar{F}^{\nu}_{\ell(i)}}{\|F^{\nu}_{i} - \bar{F}^{\nu}_{\ell(i)}\|_2 + \varepsilon_8}. \tag{5.2}$$

Why this term? Removing the category average focuses on what distinguishes items within a clothing category. This subtracts the mean feature vector of the category. It does not guarantee statistical independence from category, nor does it prove that the remaining features measure compatibility. The set $\{i : \ell(i) = a\}$ contains all catalogue items with the supplied label a, including items absent from the training bundles. The average is computed coordinate by coordinate within one modality and this one category. A category represented by one item therefore gives that item a zero residual vector. This category mean is distinct from the fitted cluster representative in Section 6.2.

Let $D_* = \mathcal{N}([D^t|D^i|D^d])$, where brackets mean concatenating feature columns. The table LD_* is a weighted sum of other items' feature rows, using exactly the entries of L above. Define

$$F_b = \mathcal{N}(D_* + 0.5\mathcal{N}(LD_*)). \tag{5.3}$$

For a candidate c, the intermediate row is $(LD_*)_c = \sum_{i \neq c} L_{ci} D_{*,i}$: a descriptor obtained from other items that share training bundles with c. It is normalized before being added to c's own residual descriptor. The resulting sum is normalized again. These two normalizations have different inputs and are both part of the definition. The three selectable feature scores are

$$r_{\text{text}}(c \mid S) = D^{t}_{c} \cdot \frac{1}{|S|} \sum_{s \in S} D^{t}_{s}, \tag{5.4}$$

$$r_{\text{image}}(c \mid S) = D^{i}_{c} \cdot \frac{1}{|S|} \sum_{s \in S} D^{i}_{s}, \tag{5.5}$$

$$\boxed{r_{\text{bridge}}(c \mid S)} \boxed{= F_{b,c} \cdot \frac{1}{|S|} \sum_{s \in S} F_{b,s}.} \tag{5.6}$$

Why this term? Text and images provide information even for items with no training appearances. The bridge adds information from historically connected items to each item's own features. Description features contribute to this bridge. Their absence as a separately weighted source is a restriction of the chosen POG_DENSE model, not a consequence of their being mathematically redundant. The bridge coefficient 0.5 is a fixed choice. Unavailable modalities and their separately weighted sources are omitted. If title features are unavailable, the agreement term is omitted and $\lambda_R = 0$.

6 Expand the category sources

6.1 Named clothing categories

The label $\ell(i)$ is the supplied `cate` field (or `cate_id` when that field is used) for item i in `item_info.json`. Distinct label values are assigned distinct integer indices. These are supplied metadata, rather than labels obtained by clustering the description vectors. The integer used to identify a category has no numerical ordering meaning; only equality of labels and the counts below are used. Let $A_{ea} = \sum_i B_{ei} \mathbf{1}[\ell(i) = a]$ count category a in bundle e. Define $H = \text{off}(A^{\top}A)$, $h_a = \sum_e A_{ea} + \varepsilon_9$, and $H_+ = \sum_a h_a$. Define the category-pair table

$$Q_{ab} = \begin{cases} [\log(H_{ab} H_+ / (h_a h_b) + \varepsilon_{12})]_+, & H_{ab} > 0, \\ 0, & H_{ab} = 0. \end{cases} \tag{6.1}$$

Let $n_a^{(1)}$ and $n_a^{(2)}$ be the numbers of training bundles containing at least one and at least two items of category a, respectively. Define $\rho_a = (n_a^{(2)} + 0.5)/(n_a^{(1)} + 1)$. Then

$$\boxed{r_{\text{cat}}(c \mid S) = \sum_{s \in S} Q_{\ell(s),\ell(c)} - 0.5 \sum_{s \in S} \mathbf{1}[\ell(s) = \ell(c)]\mathbf{1}[\rho_{\ell(s)} < 0.15](1 - \rho_{\ell(s)}).} \tag{6.2}$$

Why this term? The first part rewards category combinations seen together more often than their category counts suggest. The second discourages repeating a category that rarely appears twice in a training bundle. The counts come from training bundles; the smoothing constants $0.5, 1$, the penalty multiplier 0.5, and threshold 0.15 are prescribed choices. The ratio ρ_a estimates how frequently the category is repeated among bundles containing it; smoothing keeps it defined even for a category absent from training. For distinct categories a, b, the count H_{ab} sums $A_{ea}A_{eb}$ across training bundles: a bundle containing two items of category a and three of category b contributes six. The marginal h_a counts category occurrences, not merely bundles containing the category. Thus the category log-ratio uses the stated category counts; it must not be silently substituted by the pair-endpoint marginals used in the item PPMI table.

6.2 Clusters of items fitted from supplied features

Role in the selected dense model. This is an available source, rather than a required contribution. The saved full POG_DENSE mechanism-study recipe selects $w_{\text{soft}} = 0$ and excludes it from the fallback set $\mathcal{P}$. Its direct contribution to the final score is therefore exactly zero. The selected nonzero sources in that recipe are category, bridge, bundle, popularity and SVD. The separately fitted shared-source model selects soft with weight 0.25, together with category and bridge at 0.25 each; that restricted result does not isolate soft's incremental benefit. The formulas below define the available source so that its construction remains explicit even when its selected weight is zero.

Named clothing categories give each item one label. We additionally fit clusters of items by MiniBatch K-means, using the supplied item vectors so that items with similar descriptions can share a cluster even when their clothing labels differ. These clusters are fitted from item vectors; their indices are unrelated to clothing-category labels.

The vectors being grouped. Let the title, image, and description vectors have respectively p_t, p_i, p_d coordinates, and set $d_X = p_t + p_i + p_d$. The row F_i^ν contains the normalized supplied vector of item i for modality ν, as defined in the preceding section. For every catalogue item define

$$u_i = [F_i^t \mid F_i^i \mid F_i^d] \in \mathbb{R}^{d_X}, \qquad X_i = \frac{u_i}{\|u_i\|_2 + \varepsilon_8}. \tag{6.3}$$

Concatenation places the title coordinates first, the image coordinates next, and the description coordinates last; it does not average different modalities or different coordinates. If a modality is unavailable its block is omitted, and d_X is the sum of the remaining block lengths. These are the supplied vectors F^ν, not the category-subtracted vectors D^ν. All catalogue rows X_i, including items with $f_i = 0$, participate. Bundle completion targets are not inputs to this clustering step.

What a cluster representative is. Set $k = \max(2, \min(400, n - 1))$ and let $a \in \{1, \ldots, k\}$ index a feature cluster. Its unnormalized representative is a vector $\mu_a = (\mu_{a1}, \ldots, \mu_{a,d_X}) \in \mathbb{R}^{d_X}$ in exactly the same coordinate system as X_i. There are k such vectors, rather than one average of the whole catalogue. The MiniBatch K-means fitting criterion is

$$\min_{\mu_1,\ldots,\mu_k} \sum_{i \in \mathcal{I}} \min_{1 \le a \le k} \|X_i - \mu_a\|_2^2. \tag{6.4}$$

For fixed representatives an item is assigned to a nearest one. If the assigned items of a nonempty cluster were fixed as a set $\mathcal{A}_a$, the exact least-squares representative would have coordinates

$$\mu_{aj} = \frac{1}{|\mathcal{A}_a|} \sum_{i \in \mathcal{A}_a} X_{ij}, \qquad j = 1, \ldots, d_X. \tag{6.5}$$

Thus an average is taken over items assigned to this particular cluster, separately for each coordinate. It is not an average over feature coordinates, over every item in the catalogue, or over training bundles.

The representatives are fitted by MiniBatch K-means: repeatedly sample up to 4096 item rows, assign each sampled row to its nearest current representative, and update the representative by averaging. If N_a is the accumulated count before an update and $\mathcal{M}_a$ is the multiset of sampled rows assigned to cluster a, the ordinary update is

$$\mu_a^{\text{new}} = \frac{N_a \mu_a^{\text{old}} + \sum_{i \in \mathcal{M}_a} X_i}{N_a + |\mathcal{M}_a|}, \qquad N_a^{\text{new}} = N_a + |\mathcal{M}_a|. \tag{6.6}$$

If no sampled row is assigned to a, that update leaves it unchanged. Fitting uses pseudo-random seed zero and three candidate initializations. Initial representatives are sampled item rows, favoring rows far from representatives already selected (the k-means++ rule). Very low-count representatives may be reset to sampled rows. Thus μ_a is the vector returned by this incremental procedure; it need not equal the exact mean of the final assigned set in (6.5). The target remains the squared-distance criterion in (6.4), without assuming a global optimum. The limit of 400 clusters is a prescribed setting, independent of the number of clothing categories.

The precise definition of v_a. After fitting μ_a, normalize that vector:

$$\boxed{v_a = \frac{\mu_a}{\|\mu_a\|_2 + \varepsilon_8} \in \mathbb{R}^{d_X}.} \tag{6.7}$$

The coordinate v_{aj} is the normalized version of the representative's jth coordinate. An item-to-cluster comparison is therefore the fully specified dot product $X_i \cdot v_a = \sum_{j=1}^{d_X} X_{ij} v_{aj}$. For nonzero vectors it approximates their cosine similarity, with the stated numerical stabilizers. If $\mu_a = 0$, then $v_a = 0$ and this dot product is zero for every item. For example, two assigned rows $(1, 0)$ and $(0, 1)$ have exact mean $(1/2, 1/2)$ and normalized direction approximately $(1/\sqrt{2}, 1/\sqrt{2})$. Their two coordinates remain separate throughout the calculation.

From a representative to an item's cluster weights. The cluster memberships used in the following score are soft: a row is compared with every representative, even though nearest-representative assignments were used to fit them. Set $\tau_s = 0.05$ and define

$$t_{ia} = X_i \cdot v_a / \tau_s, \qquad J_{ia} = \frac{\exp(t_{ia} - \max_b t_{ib})}{\sum_b \exp(t_{ib} - \max_b t_{ib}) + \varepsilon_9}. \tag{6.8}$$

Both the maximum and sum range over the k feature clusters. The exponential assigns larger weight to a more aligned representative; τ_s controls how concentrated these weights are. Subtracting the maximum avoids large exponentials without changing the ratio when the denominator stabilizer is omitted. Because of that stabilizer the row sum of J is slightly less than one. No clothing label or bundle overlap enters this item-to-cluster comparison. Bundle membership enters next, when these fitted weights are aggregated within training bundles. Let $T = BJ$, so $T_{ea} = \sum_i B_{ei} J_{ia}$ is the total membership of items in training bundle e in cluster a. Define cluster-pair counts, catalogue cluster masses, and their total by

$$H^s = \left[\operatorname{off} \left(T^\top T - \sum_e \sum_{i : B_{ei} = 1} J_i^\top J_i \right) \right]_+, \qquad a_u = \sum_i J_{iu}, \qquad A_s = \sum_u a_u. \tag{6.9}$$

Finally, define $Q^s_{uv} = [\log(H^s_{uv} A_s/(a_u a_v + \varepsilon_9) + \varepsilon_{12})]_+$. The raw score is

$$\boxed{r_{\text{soft}}(c \mid S) = \sum_{s \in S} \sum_{u=1}^{k} \sum_{v=1}^{k} J_{cu} Q^s_{uv} J_{sv}.} \tag{6.10}$$

Each summand combines the candidate's weight in cluster u, the learned association between clusters u, v, and an observed item's weight in cluster v. Thus cluster relations are transferred back to item scores.

Why this term? An item can partly belong to several feature clusters. Their training-bundle relationships may capture combinations missed by a single clothing label. In fact, before the diagonal is removed, the table being counted is

$$T^\top T - \sum_e \sum_i B_{ei} J_i^\top J_i = \sum_e \sum_{i \neq j} B_{ei} B_{ej} J_i^\top J_j.$$

This is a sum over distinct items in the same training bundle. The subtraction therefore removes an item's contribution paired with itself; otherwise a feature cluster could appear to complement itself merely because it describes the same item twice.

Relation to earlier methods. Let x_S denote the catalogue indicator vector of the observed set. The entire soft score vector can be written as

$$r_{\text{soft}}(\cdot \mid S) = J Q^s J^\top x_S.$$

It maps observed items to cluster memberships, applies cluster relations, and maps back to candidate items. The resulting item matrix has rank at most k; it is not necessarily positive semidefinite. Before removal of the cluster diagonal, the distinct-item count table is exactly $J^\top C J$, because $C = B^\top B - \text{diag}(f_i)$. Thus this source shares pair evidence through feature clusters; it does not retain additional triple-occurrence information.

Soft item clusters are established in recommendation. For example, Puthiya Parambath and Chawla (2020), *Simple and effective neural-free soft-cluster embeddings for item cold-start recommendations*, derive item memberships from enriched side information using nonnegative matrix factorization, then fit user vectors for prediction. Their memberships and prediction rule differ from the count-derived Q^s used here. This connection supports treating soft clustering as an established ingredient. It does not establish that this paper used the exact self-item subtraction and log-ratio construction above, or that those details are new.

7 Expand the pair-of-known-items term r_R

This optional score asks: does a candidate receive support from two different observed items, and how different are those two supporting items? It does not require the candidate and both supporting items to occur in the same training bundle. The saved full POG_DENSE mechanism-study recipe selects $\lambda_R = 0$, so this source has zero contribution to that final score. We nevertheless define it completely as an available source.

7.1 Define the inputs and distinguish the two uses of D

The catalogue $\mathcal{I}$ contains n items. The candidate is $c \notin S$; $s, t \in S$ are observed items, and $e \in \{1, \ldots, m\}$ indexes training bundles. B_{ei} is one when bundle e contains item i. Recall $f_i = \sum_e B_{ei}$ and $d_e = \sum_i B_{ei}$.

Earlier sections use two different objects with the letter D:

$$D = \operatorname{diag}((f_i + 5)^{-1/2}) \in \mathbb{R}^{n \times n}, \qquad D^t \in \mathbb{R}^{n \times p_t}.$$

The first is a diagonal table of frequency weights. The second is a table of residual title vectors, with p_t title coordinates per item; its superscript t means *title*, not transpose and not the observed item indexed by t. To avoid confusing them in this section, write $\widehat{t}_i = D_i^t$ for item i's residual title vector. Its full definition is

$$F_i^t = \frac{R_i^t}{\|R_i^t\|_2 + \varepsilon_8}, \qquad \mathcal{I}_a = \{i \in \mathcal{I} : \ell(i) = a\}, \tag{7.1}$$

$$\bar{F}_a^t = \frac{1}{|\mathcal{I}_a|} \sum_{i \in \mathcal{I}_a} F_i^t, \qquad \widehat{t}_i = \frac{F_i^t - \bar{F}_{\ell(i)}^t}{\|F_i^t - \bar{F}_{\ell(i)}^t\|_2 + \varepsilon_8}. \tag{7.2}$$

Here R_i^t is the supplied title vector, $\ell(i)$ is the supplied clothing category, and $\mathcal{I}_a$ contains catalogue items in category a. The mean is taken within that category, coordinate by coordinate. A zero residual stays zero. Thus $\widehat{t}_i \cdot \widehat{t}_j$ compares two items' deviations from their respective category-average title vectors.

7.2 Compute support from each observed item

Define W_{ci} as the weighted co-membership of candidate c and item i:

$$W_{ci} = \begin{cases} \dfrac{1}{\sqrt{f_i + 5}} \displaystyle\sum_{e=1}^{m} \frac{B_{ec} B_{ei}}{\sqrt{d_e + 1}}, & c \neq i, \\ 0, & c = i. \end{cases} \tag{7.3}$$

Equivalently $W = \operatorname{off}(B^\top V B D)$ with $V = \operatorname{diag}((d_e + 1)^{-1/2})$. The scalar formula shows exactly where the diagonal frequency table D enters: it discounts the observed item's frequency f_i. W is different from the symmetric propagation table L defined earlier.

Set $\eta = 0.15$ and define

$$q_{cs} = W_{cs} + \eta[\widehat{t}_c \cdot \widehat{t}_s]_+, \qquad A_c = \sum_{s \in S} q_{cs}. \tag{7.4}$$

The scalar $q_{cs} \geq 0$ is the support given to candidate c by observed item s. It adds recorded co-membership to positive residual-title similarity. A_c is total support across S. These scores are not probabilities; in particular, A_c need not equal one. The positive part discards a negative residual-title dot product rather than turning it into negative support.

7.3 Discount two supporting items when they resemble each other

For distinct observed items s, t, define

$$b_{st} = \mathbf{1}[\ell(s) = \ell(t)], \qquad h_{st} = [\widehat{t}_s \cdot \widehat{t}_t]_+. \tag{7.5}$$

Here b_{st} is one for the same category and zero otherwise; h_{st} is positive residual-title similarity. Set $\omega = 0.5$ and form

$$G_{st} = \omega b_{st} + (1 - \omega) h_{st}, \qquad K_{st} = [1 - G_{st}]_+, \qquad K_{ss} = 0. \tag{7.6}$$

In exact arithmetic, both b_{st} and h_{st} lie in $[0, 1]$, so their convex combination does too. For distinct items the weight can therefore be written

$$\boxed{K_{st} = \tfrac{1}{2}(1 - b_{st}) + \tfrac{1}{2}(1 - h_{st}).} \tag{7.7}$$

This is the average of category dissimilarity and positive residual-title dissimilarity, as defined here. It is a prescribed discount multiplier on the product of two support scores; its exact contribution is expanded below. Setting $K_{ss} = 0$ prevents a seed from being counted as agreeing with itself.

Written numerically the category summand is

$$\underbrace{0.5}_{\text{weight}} \times \underbrace{\mathbf{1}[\ell(s) = \ell(t)]}_{\text{either 0 or 1}}.$$

This is 0.5 *times an indicator*, not the decimal number 0.51. Its value is either 0.5 or zero. For example, the same category with $h_{st} = 0$ gives $G_{st} = 0.5$ and $K_{st} = 0.5$; different categories with $h_{st} = 0$ give $G_{st} = 0$ and $K_{st} = 1$. The same category with h_{st} near one gives K_{st} near zero. These are consequences of the stated rule, not assertions that category and residual-title similarity exhaust all possible forms of redundancy. The modeling assumption is that equally strong candidate support should receive more agreement weight when its two supporting items differ by these measures. The formula implements that assumption. It does not establish statistical independence of the two clues, or estimate their information gain from data.

7.4 Combine distinct pairs of support

Set $\tau_R = 0.05$. The raw agreement score is

$$\boxed{r_R(c \mid S) = \frac{\sum_{s \in S} \sum_{t \in S} q_{cs} K_{st} q_{ct}}{A_c + \tau_R}.} \tag{7.8}$$

Each summand has three factors: support from s, the weight for counting s, t as different clues, and support from t. It is positive only when both seeds support the candidate and $K_{st} > 0$. Both orders of a pair are included. Since $K_{st} = K_{ts}$ and $K_{ss} = 0$, write the same score as a sum over unordered distinct pairs:

$$r_R(c \mid S) = \sum_{\{s,t\} \subseteq S} \underbrace{\frac{2 q_{cs} q_{ct}}{A_c + \tau_R} K_{st}}_{\text{contribution of this pair to the raw score}}. \tag{7.9}$$

Each pair appears once in this sum; the factor two accounts for the two orders in (7.8). With the support scores and query fixed, $K_{st} = 1$ retains the full pair contribution, $K_{st} = 0.5$ halves it, and $K_{st} = 0$ removes it. This specifies precisely what the discount does.

This additive pair contribution is not the change in score caused by adding item t to S. Adding an item also changes A_c and introduces pairs with other observed items. It is also not an additive contribution to the final standardized score: catalogue standardization follows this raw calculation. For exactly two observed items,

$$r_R(c \mid \{s, t\}) = \frac{2 q_{cs} q_{ct} K_{st}}{q_{cs} + q_{ct} + \tau_R}.$$

For illustration, hold $q_{cs} = 0.2$ and $q_{ct} = 0.3$ fixed. Then $r_R = 2(0.2)(0.3)K_{st}/(0.2 + 0.3 + 0.05) \simeq 0.21818 K_{st}$:

Same category?	h_{st}	K_{st}	Raw pair contribution
No	0	1	0.21818
Yes	0	0.5	0.10909
No	0.8	0.6	0.13091
Yes	0.8	0.1	0.02182

These are conditional calculations at the stated illustrative support values, not measured dataset scores. With one seed the score is zero, and for a larger set any supported distinct pair can contribute.

Why divide by total support? If every distinct-pair weight were one, the numerator would be $A_c^2 - \sum_s q_{cs}^2$. More generally, for $A_c > 0$ define $p_{cs} = q_{cs}/A_c$. Then (7.8) becomes

$$r_R(c \mid S) = \frac{A_c^2}{A_c + \tau_R} \sum_{s \neq t} p_{cs} K_{st} p_{ct}.$$

The sum measures how support is spread across different clues. The leading factor measures its strength: approximately A_c when $A_c \gg \tau_R$, and approximately A_c^2/τ_R when $A_c \ll \tau_R$. The denominator therefore limits quadratic growth at high support and suppresses very weak support. If $A_c = 0$, all q_{cs} are zero and the score is zero.

7.5 Which constants are chosen, and which are fitted?

Constant	Value	Role and origin
η	0.15	Fixed multiplier on positive title similarity in q_{cs}; not estimated by the fitting procedure.
ω	0.5	Fixed equal weighting of category equality and residual-title similarity in G_{st}. Its complement is also 0.5.
τ_R	0.05	Fixed positive support scale in the denominator; not a fitted variance or numerical epsilon.
λ_R	$\{0, 0.2, 0.35\}$	Outer coefficient selected on validation rankings after standardizing r_R; selected value is zero in the saved full dense recipe.

The first three values are prescribed modeling choices. Their roles have mathematical explanations, but the equations do not determine their exact numerical values and the recorded procedure does not tune them separately. The value 0.5 expresses an equal-weight convention; 0.15 sets the relative size of descriptor support; 0.05 sets the transition between weak and strong support. No evidence in the recorded procedure establishes these particular numbers as optimal. Validation tests the usefulness of the resulting fixed source through λ_R, rather than estimating all its internal constants. The offsets 5 and 1 in W are the same fixed frequency and bundle-size smoothing choices already defined for the direct bundle score.

Finally standardize (7.8) using (2.3) to obtain z_R. The selected contribution enters Y and P as $\lambda_R z_R$, followed by the final fallback mixture. The unstandardized r_R is nonnegative, but z_R can be negative for a candidate below the catalogue-average agreement. If title vectors are unavailable this source is omitted and $\lambda_R = 0$.

8 Where the fixed settings come from

The expansion of θ is now complete except for how $w_h, \alpha, \lambda_R, s_0, \kappa$ and $\mathcal{P}$ are chosen. They are selected on separate examples with known hidden items, then held fixed when scoring new requests. The supplied validation inputs and targets are `bi_valid_input.txt` and `bi_valid_gt.txt`; the corresponding test files are `bi_test_input.txt` and `bi_test_gt.txt`. Their observed/hidden partitions are retained as supplied. Test targets are used only for evaluation and do not enter these choices or the scoring formulas. In particular, α could mathematically be any value in $[0, 1]$, and λ_R any nonnegative value. The finite sets below are the search candidates chosen for this method, so that fitting uses finitely many evaluations. They are not the only admissible values, and no claim of optimal spacing or exhaustive optimization follows. The same distinction applies to the weight grid, tolerances, four-pass limit and clipping bounds. Their numerical values specify

this procedure; a mathematical justification for these exact values is not supplied by the scoring equations.

Weights and mixture. Visit the sources in the order cat, soft, text, image, bridge, bundle, propagated, pop, cooc, pmi, svd. Starting at zero, try each weight in $\{0, 0.25, 0.5, 1, 1.5\}$, skipping the all-zero weight vector and using Y_0 and keeping changes that improve mean NDCG@20 by more than 2×10^{-4}, for at most four passes. For request j, let G_j be its nonempty hidden-item set, let $\pi_j(r)$ be its rth ranked non-known item, and let $K_j = \min(20, n - |S_j|)$. Define

$$\mathrm{NDCG}_{20,j} = \frac{\sum_{r=1}^{K_j} \mathbf{1}[\pi_j(r) \in G_j] / \log_2(r+1)}{\sum_{r=1}^{\min(20,|G_j|)} 1/\log_2(r+1)}.$$

Mean NDCG@20 is its average across held-out requests. Try removing bridge and soft-cluster sources if doing so loses at most 3×10^{-4}, retaining at least one positive-weight source. Choose $\mathcal{P}$ from positive-weight category/soft-cluster/text/image/bridge sources, falling back to all available ones if necessary. Then compare $\alpha \in \{0, 0.5, 1\}$ and $\lambda_R \in \{0, 0.2, 0.35\}$, retaining improvements exceeding 5×10^{-5}.

Why this term? The method learns which evidence is useful for the completion task without assuming that every source deserves equal importance.

Scale. Across all catalogue items and held-out requests, let sd denote sample standard deviation. With the selected weights and mixture, first estimate

$$s_* = \frac{\mathrm{sd}(Y) + \varepsilon_9}{\mathrm{sd}(P) + \varepsilon_9}. \tag{8.1}$$

Why this term? This aligns the overall size of variations in the two estimates before mixing them.

History adjustment. Let there be v held-out requests indexed by j. For each item with $f_i > 0$, define $x_i = 1/f_i$ and $e_i = v^{-1} \sum_j (Y(i \mid S_j) - s_* P(i \mid S_j))^2$. Let $\bar{x}, \bar{e}$ be their means across such items. Fit a straight line to these values:

$$b_* = \frac{\sum_i (x_i - \bar{x})(e_i - \bar{e})}{\sum_i (x_i - \bar{x})^2}, \qquad a_* = \bar{e} - b_* \bar{x}, \tag{8.2}$$

$$\kappa_* = \mathrm{clip}\left(\frac{\max(b_*, 10^{-4})}{\max(a_*, 10^{-4})}, 0.05, 50 \right). \tag{8.3}$$

If fewer than two items have history, or the denominator is at most 10^{-12}, use only $\kappa = 0$. Otherwise compare $\kappa \in \{0, 0.05, 0.5, 1, \kappa_*, 5\}$ with $s_0 \in \{s_*, 1\}$, retaining NDCG@20 improvements greater than 5×10^{-5}.

Why this term? Let θ_i^* be an unobserved suitability, and $\mu_i = s_* P(i \mid S)$. Assume $\theta_i^* = \mu_i + \xi_i$ and $Y_i = \theta_i^* + \eta_i$, with independent, zero-mean normal errors of variances σ_0^2 and σ_e^2 / f_i, respectively. For $f_i > 0$ these assumptions imply

$$\mathbb{E}[(Y_i - \mu_i)^2 \mid f_i] = \sigma_0^2 + \frac{\sigma_e^2}{f_i}, \qquad \mathbb{E}[\theta_i^* \mid Y_i, \mu_i, f_i] = \frac{f_i Y_i + \kappa \mu_i}{f_i + \kappa}, \quad \kappa = \frac{\sigma_e^2}{\sigma_0^2}.$$

This explains both the fitted line and the rational weights in equation (1.1). The assumptions are a model of score errors, not a consequence of the observed counts. The slope and intercept estimate the two variances; positivity bounds and clipping turn their ratio into one search candidate. Held-out ranking quality chooses the final setting. The squared residuals compare Y and $s_* P$, not either score with a binary item label.

Producing the completion. Standardize the selected raw scores with (2.3), form Y and P, and evaluate (1.1). Exclude S and return the K highest scores, breaking exact ties by increasing item ID. A source with zero weight and no fallback role does not enter the calculation.